Assignment 5

1 Grid Fees: Introduction

In Assignments 1 and 2 we formulated the household battery scheduling problem as two integer linear programmes—the BO-MIP and TO-MIP—following Elgersma et al. (2024). Those formulations maximise net revenue from grid exchange without accounting for network charges. In practice, however, Dutch households on dynamic contracts pay a grid fee for every kilowatt-hour exchanged with the distribution network ACM (2023). We extend both formulations with a symmetric, proportional fee $\varphi \geq 0$ (€/kWh) levied on imports and exports, representing the volumetric transport tariffs applied under the *Tarievencode Elektriciteit* ACM (2023). Section 2 develops the extension, Section 3 defines battery utilisation metrics, Section 4 presents results and discusses their implications, and Section 5 concludes.

2 Grid Fees: Model Extension

The base objective maximises net grid revenue $\sum_{t\in\mathcal{T}} \lambda_t (g_t^{\text{out}} - g_t^{\text{in}})\,\Delta$. The total fee incurred in period t is $\varphi\,(g_t^{\text{in}} + g_t^{\text{out}})\,\Delta$. Subtracting this and regrouping yields

$$\max \sum_{t\in\mathcal{T}} \Big[\underbrace{-(\lambda_t + \varphi)}_{\text{eff. buy price}}\; g_t^{\text{in}} + \underbrace{(\lambda_t - \varphi)}_{\text{eff. sell price}}\; g_t^{\text{out}} \Big]\,\Delta. \tag{1}$$

The fee creates a wedge of 2φ between the effective buying and selling prices. No other changes to either formulation are required: the power balance already decomposes grid exchange into g_t^{in} and g_t^{out}, and all constraints from Assignments 1–2 remain unchanged. Because $\lambda_t + \varphi > 0$ for all periods and fee levels in our dataset, the objective drives complementarity ($g_t^{\text{in}} \cdot g_t^{\text{out}} = 0$) in any optimal solution without explicit binary enforcement Wolsey (1998). Since the constraint sets are unmodified, the TO-MIP LP relaxation still describes the convex hull of integer-feasible solutions (Elgersma et al., 2024, Theorem 1), and both formulations produce identical optimal schedules at every fee level.

3 Grid Fees: Battery Utilisation Metrics

To assess how the grid fee alters battery behaviour, we define three metrics and one derived decomposition. Each captures a qualitatively different aspect of the battery's response: physical wear, network impact, and energy self-sufficiency.

M1—Equivalent full cycles. The number of equivalent full cycles measures the total energy discharged, normalised by the usable battery capacity:

$$N_{\text{cyc}} = \frac{\sum_{t\in\mathcal{T}} p_t^{\text{dis}}\,\Delta}{S^{\max} - S^{\min}}. \tag{2}$$

This is the standard metric for cycle-ageing assessment in lithium-ion batteries Xu et al. (2018): each full cycle degrades the battery, so N_{cyc} serves as a proxy for wear. A decline in N_{cyc} as the fee rises indicates that the battery is cycling less intensively, which has the side benefit of extending its calendar and cycle lifetime. However, as we will see in the results, cycles do not fall to zero even at high fees—the battery continues to cycle for self-consumption rather than arbitrage.

M2—Total grid exchange. The total volume of electricity flowing between the household and the grid is

$$G_{\text{total}} = \sum_{t \in \mathcal{T}} \left(g_t^{\text{in}} + g_t^{\text{out}} \right) \Delta. \tag{3}$$

This metric is relevant from two perspectives. For the household, the total fee payment equals $\varphi \cdot G_{\text{total}}$, so it directly determines the fee burden. For the distribution system operator, G_{total} reflects the household's contribution to network loading: reducing it alleviates congestion on local feeders, which is one of the primary motivations for volumetric grid fees Sioshansi (2010).

M3—Self-consumption rate. Let $\sigma_t = \max(0, -d_t)$ denote the local solar surplus in period t. The self-consumption rate is

$$\text{SCR} = 1 - \frac{\sum_{t \in \mathcal{T}} \min(\sigma_t, g_t^{\text{out}}) \, \Delta}{\sum_{t \in \mathcal{T}} \sigma_t \, \Delta}, \tag{4}$$

whenever the denominator is positive; we set SCR = 1 if total solar surplus is zero. The numerator in the fraction represents solar energy that is exported rather than consumed locally; subtracting its share from unity yields the fraction consumed on-site. This metric is central to Dutch energy policy: as the *salderingsregeling* (net-metering scheme) is phased out, households are increasingly incentivised to maximise self-consumption Schill et al. (2017); Luthander et al. (2015).

Derived metric—Behavioural cost. We decompose total profit loss into a direct fee burden and a behavioural cost representing profitable trades the household forgoes to avoid paying the fee:

$$C_{\text{behav}}(\varphi) = \underbrace{(\pi_0 - \pi_\varphi)}_{\text{total profit loss}} - \underbrace{\varphi \cdot G_{\text{total}}(\varphi)}_{\text{direct fee burden}}, \tag{5}$$

where π_φ denotes the optimal objective value at fee level φ. If the household did not adjust its schedule at all, the profit loss would equal $\varphi \cdot G_{\text{total}}(0)$ and C_{behav} would be zero. A positive behavioural cost means the household *does* adjust—forgoing otherwise profitable arbitrage to reduce fee exposure—but the adjustment itself comes at a cost because those trades were profitable before the fee was introduced. This decomposition separates the mechanical fiscal transfer from the efficiency loss caused by distorted incentives.

4 Grid Fees: Results and Discussion

Both formulations were solved over the full 7 895-period dataset under the parameters in Table 3, for six fee levels from 0 to 0.05 €/kWh, using PuLP with the CBC solver. All instances reached optimality, and the BO-MIP and TO-MIP produced identical solutions at every fee level. Table 1 summarises the results.

Objective decline versus battery value. The objective drops from €32.26 at $\varphi = 0$ to −€156.82 at $\varphi = 0.05$. A superficial reading suggests that the fee destroys the battery's value, but this conflates two distinct quantities. The objective is the household's *total* net grid cost; the battery's *incremental* value is the improvement over a no-battery baseline. Without a battery, the household passively imports 1 847 kWh and exports 1 567 kWh, incurring a net grid cost of −€138.71 at $\varphi = 0$. The battery improves this by €170.96. Table 2 and Figure 1 show that this incremental value remains stable across all fee levels, following a U-shaped curve with a minimum of €143.98 at $\varphi = 0.02$ and recovering to €152.59 at $\varphi = 0.05$.

Table 1: Grid-fee sweep results (7 895 periods, BO-MIP and TO-MIP identical).

φ (€/kWh)	Objective (€)	Fee paid (€)	C_{behav} (€)	Import (kWh)	Export (kWh)	G_{total} (kWh)	N_{cyc}	SCR (%)
0.00	32.26	0.00	0.00	3 534	2 989	6 523	548	36.2
0.01	−22.73	46.53	8.46	2 575	2 078	4 653	449	49.8
0.02	−63.00	71.48	23.78	2 018	1 556	3 574	376	54.2
0.03	−96.43	95.63	33.06	1 819	1 369	3 188	352	57.1
0.04	−127.31	120.06	39.51	1 723	1 279	3 001	339	58.5
0.05	−156.82	145.46	43.62	1 675	1 234	2 909	332	59.4

Table 2: Battery incremental value at each fee level.

φ (€/kWh)	No-battery (€)	With battery (€)	Battery value (€)
0.00	−138.71	32.26	170.96
0.01	−172.85	−22.73	150.11
0.02	−206.99	−63.00	143.98
0.03	−241.13	−96.43	144.69
0.04	−275.27	−127.31	147.96
0.05	−309.41	−156.82	152.59

Arbitrage-to-self-consumption role shift. The U-shape in Figure 1 is explained by a role shift. At $\varphi = 0$, the battery creates value through arbitrage: it imports 3 534 kWh (nearly double the passive 1 847 kWh), adding large volumes to serve buy-low-sell-high round trips. At $\varphi = 0.05$, import is 1 675 kWh—*below* the passive baseline—and export is 1 234 kWh, also below the passive 1 567 kWh. The battery no longer adds exchange volume; it *reduces* it by storing solar surplus for later use, avoiding both the export and a subsequent reimport. For low fees, the declining arbitrage value dominates and battery value falls. For high fees, each kWh of self-consumption avoids 2φ in fees, so the self-consumption value rises and eventually outweighs the lost arbitrage. The minimum near $\varphi = 0.02$ marks the crossover between these two roles.

Behavioural threshold. The steepest adjustments concentrate between $\varphi = 0$ and 0.02: grid exchange falls by 45%, import by 43%, and equivalent cycles by 31%. The mechanism is the round-trip profitability condition: for a charge–discharge cycle with buy price λ_{buy} and sell price λ_{sell}, the profit per kWh stored is $\pi_{\text{rt}} = (\lambda_{\text{sell}} - \varphi)\,\eta^{\text{D}} - (\lambda_{\text{buy}} + \varphi)/\eta^{\text{C}}$, requiring a spread that exceeds the efficiency loss plus $\varphi(1/\eta^{\text{D}} + 1/\eta^{\text{C}})$. With $\eta^{\text{C}} = \eta^{\text{D}} = 0.95$, each €0.01 of fee raises the minimum spread by ∼0.021 €/kWh, eliminating a large class of marginal round trips. Beyond $\varphi = 0.03$, the remaining exchange is predominantly obligatory and responds weakly to further increases.

Self-consumption. Figure 2 shows the SCR rising from 36% to 59%, a 23 percentage-point increase. The largest gain (+14 pp) occurs at $\varphi = 0.01$; subsequent increments yield diminishing returns as the remaining export consists of surplus the battery cannot absorb. The 332 equivalent cycles that persist at $\varphi = 0.05$ are largely self-consumption cycles rather than arbitrage cycles: the battery charges from midday solar surplus and discharges to serve evening demand, avoiding grid exchange in both directions Schill et al. (2017); Weniger et al. (2014).

Role of negative prices. The dataset contains 547 periods with negative spot prices. During these hours, the effective buy price $\lambda_t + \varphi$ can remain negative, meaning the household is paid to charge the battery even after the fee. This creates profitable cycling opportunities that persist at high fee levels and partly explains why equivalent cycles do not decline further beyond $\varphi = 0.03$.

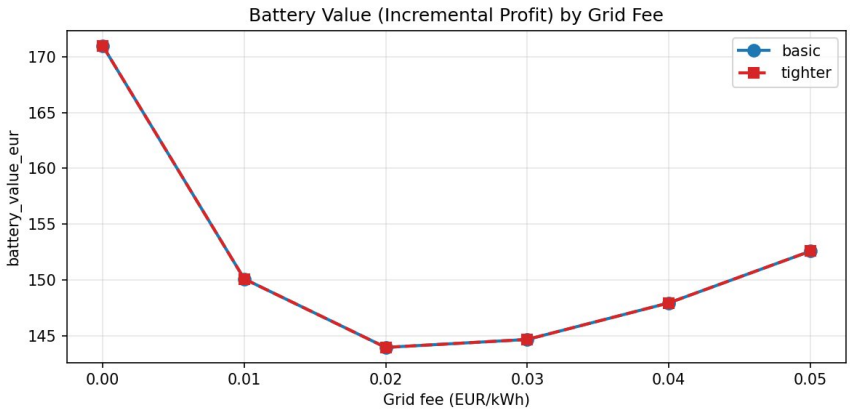

Figure 1: Battery incremental value by fee level. The U-shaped curve reflects the transition from arbitrage-dominated value (declining with φ) to self-consumption-dominated value (increasing with φ).

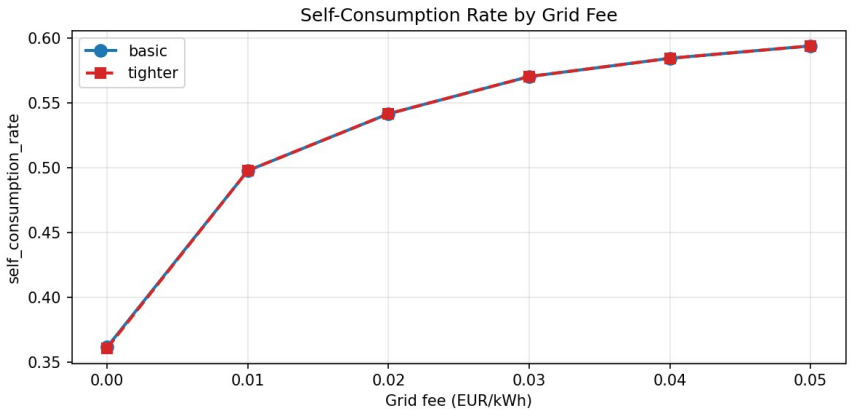

Figure 2: Self-consumption rate by fee level.

Policy implications. A symmetric volumetric fee effectively reduces grid exchange (-55% at $\varphi = 0.05$) and promotes self-consumption ($+23$ pp), aligning household incentives with network objectives Sioshansi (2010). Crucially, however, the battery retains substantial incremental value at all fee levels (Table 2), suggesting that volumetric fees alone cannot discourage battery adoption. The fee's main effect is to redirect battery use from arbitrage toward self-consumption—a shift that regulators may welcome, since arbitrage increases grid loading while self-consumption reduces it. An asymmetric or time-varying tariff could further refine these incentives, motivating the extensions in Assignment 7.

5 Grid Fees: Conclusion

We extended both formulations with a symmetric proportional grid fee that modifies only the objective function. The parametric sweep reveals that although the objective turns negative at $\varphi = 0.01$, the battery's incremental value remains stable between €144 and €171, following a U-shaped curve that reflects a transition from arbitrage-dominated to self-consumption-dominated value. The steepest behavioural response occurs between $\varphi = 0$ and 0.02, where grid exchange falls by 45% and self-consumption rises from 36% to 54%. At high fees, import volumes fall below the no-battery passive baseline, confirming the battery's transition to a self-consumption device. Both formulations produce identical solutions at all fee levels, confirming that the TO-MIP

Table 3: Reference parameters.

Symbol	Value	Description
S_0	2 kWh	Initial state of charge
$S^{\min}$, $S^{\max}$	0.5, 5 kWh	State-of-charge bounds
P^{C}, P^{D}	2 kW	Power limits
η^{C}, η^{D}	0.95	Efficiencies
Δ	1 h	Time-step duration
φ	0–0.05 €/kWh	Grid fee (swept)

tightness properties are preserved.

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